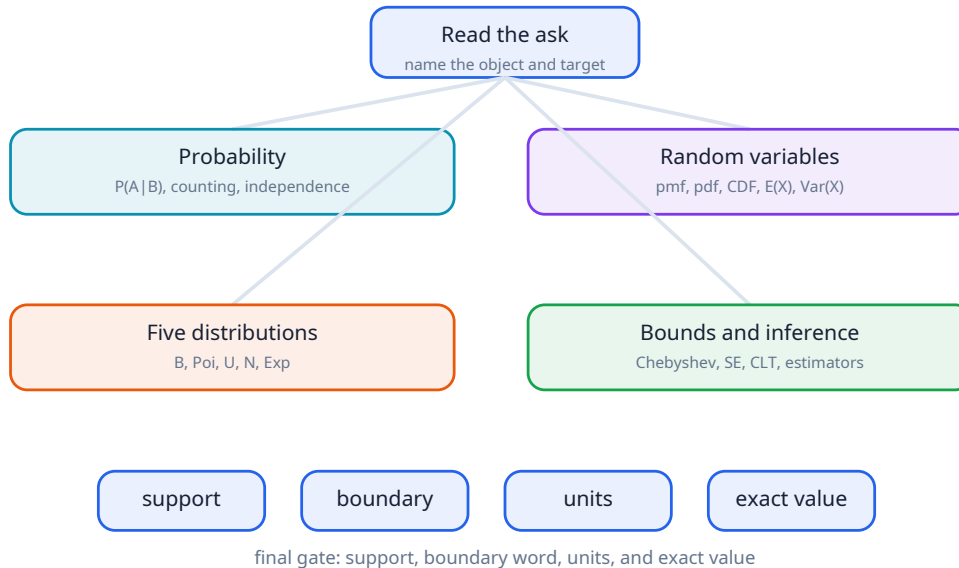


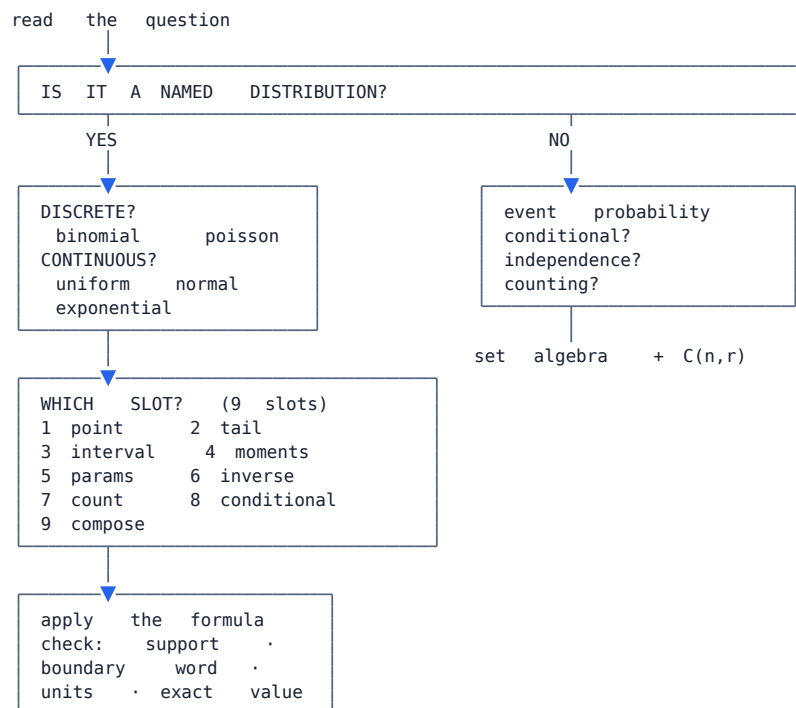
01 CHEATSHEET: everything, one pass

THE WHOLE PAPER AS ONE FLOWCHART

Formula landscape: classify first, then compute

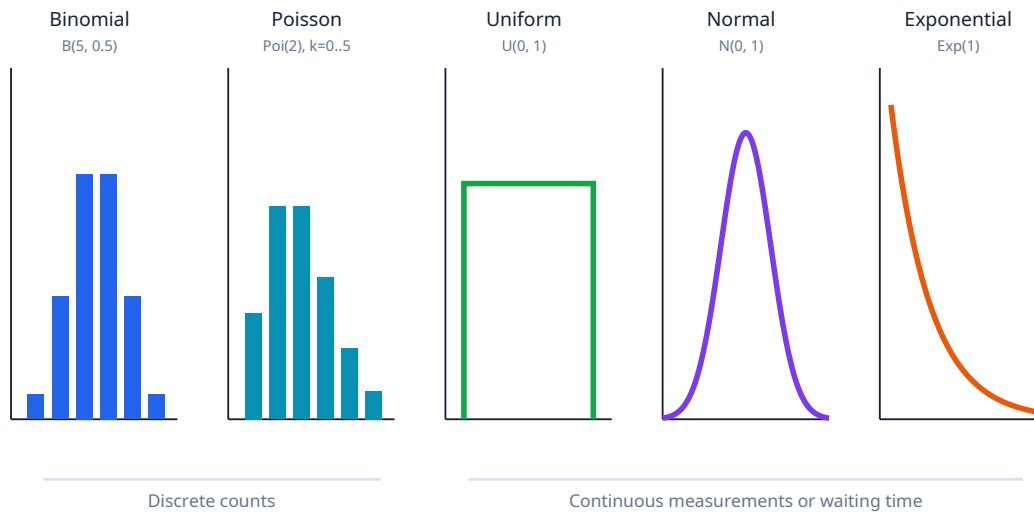


Formula landscape and final checks



THE FIVE DISTRIBUTIONS AT A GLANCE

Five distribution shapes



Five parameterized distribution shapes

DISCRETE

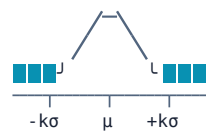
BINOMIAL np, npq
POISSON λ, λ
(mean=var!)

CONTINUOUS

UNIFORM $(a+b)/2, (b-a)^2/12$
NORMAL μ, σ^2
EXPONENTIAL $1/\lambda, 1/\lambda^2$

CHEBYSHEV, ONE PICTURE

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2$$



■ = the tail event (bounded)

k	tail	centre
2	.25	.75
3	.111	.889
4	.0625	.9375

Closed book paper. This page is the whole exam in formulas. Read it daily, last thing at night.

A. Probability

$P(S)=1$ $P(A) \geq 0$ disjoint additivity on S
 $P(A') = 1 - P(A)$
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 disjoint: $P(A \cup B) = P(A) + P(B)$
 $P(A|B) = P(A \cap B)/P(B)$, $P(B) > 0$
 $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$
 independent: $P(A \cap B) = P(A)P(B)$
 $(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$
 $n! / 0! = 1$ $C(n,r) = n! / (r!(n-r)!)$ $P(n,r) = n! / (n-r)!$ $C(n,r) = C(n,n-r)$

B. Random variables, pmf, pdf, cdf

discrete $p(x) \geq 0$, $\sum p(x) = 1$, $P(X \in A) = \sum$
 continuous $f(x) \geq 0$, $\int f \, dx = 1$, $P(a \leq X \leq b) = \int_a^b f \, dx$, $P(X=x)=0$
 cdf $F(x) = P(X \leq x)$, non decreasing, $F(-\infty)=0$, $F(\infty)=1$
 $P(a < X \leq b) = F(b) - F(a)$
 discrete $p(x) = F(x) - F(x-)$ (the jump)
 continuous $F(x) = \int_{-\infty}^x f \, dx$, $f(x) = F'(x)$
 integer rv $P(a \leq X \leq b) = F(b) - F(a-1) \leftarrow$ the $a-1$, the commonest slip

C. Expectation

discrete $E(X) = \sum x p(x)$ continuous $E(X) = \int x f \, dx$
 $E[h(X)] = \sum h(x)p(x)$ or $\int h(x) f \, dx$
 $E(aX+b) = aE(X)+b$ $E(c) = c$ $E(X+Y) = E(X)+E(Y)$ ALWAYS
 $E(XY) = E(X)E(Y)$ only if INDEPENDENT
 expected count: out of N items, expected = $N \cdot P$

D. Variance

$\text{Var}(X) = E[(X-\mu)^2] = E(X^2) - [E(X)]^2$
 $\text{sd} = \sqrt{\text{Var}}$
 $\text{Var}(aX+b) = a^2 \text{Var}(X)$ $\text{sd}(aX+b) = |a| \text{sd}(X)$ $\text{Var}(c)=0$
 $\text{Var}(X+Y) = \text{Var}(X)+\text{Var}(Y)$ independent only
 $\text{Var}(X-Y) = \text{Var}(X)+\text{Var}(Y)$ independent only, the MINUS still ADDS
 $\text{Var}(\bar{X}) = \sigma^2 / n$

E. Chebyshev

$P(|X-\mu| \geq k \sigma) \leq 1/k^2$ $k > 0$
 $P(|X-\mu| \geq \epsilon) \leq \sigma^2/\epsilon^2$
 $P(|X-\mu| < k \sigma) \geq 1 - 1/k^2$ complement form
 $k=2 \rightarrow 3/4$ $k=3 \rightarrow 8/9$ $k=4 \rightarrow 15/16$ $k=1 \rightarrow 1$, useless
 k is in standard deviations: $k = \epsilon/\sigma$
 no distribution assumed, only finite μ and variance
 always an UPPER bound on the tail, so actual $\leq$ bound, say so when asked

F. The five distributions

BINOMIAL $X \sim B(n, p)$ $q = 1 - p$
 $P(X=x) = C(n, x) p^x q^{(n-x)}$ $E = np$ $\text{Var} = npq$
 needs: 2 outcomes, finite n , independent trials, p constant
 POISSON $X \sim \text{Poi}(l)$
 $P(X=x) = e^{-l} l^x / x!$ $E = l$ $\text{Var} = l$ (mean = variance)
 $l = np$ for the rare-event approximation, n large p small
 UNIFORM $X \sim U(a, b)$
 $f = 1/(b-a)$ $E = (a+b)/2$ $\text{Var} = (b-a)^2/12$ $F = (x-a)/(b-a)$
 probability = length ratio
 NORMAL $X \sim N(\mu, \sigma^2)$ (second slot is VARIANCE)
 $Z = (X - \mu)/\sigma$ $X = \mu + Z \sigma$
 $P(a < X < b) = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$
 $\Phi(-z) = 1 - \Phi(z)$ $P(Z > z) = 1 - \Phi(z)$
 landmarks: 68.27 within 1s, 95.45 within 2s, 99.73 within 3s
 EXPONENTIAL $T \sim \text{Exp}(l)$ (l is a RATE)
 $f = l e^{-l t}$ $F = 1 - e^{-l t}$ $P(T > t) = e^{-l t}$
 $E = 1/l$ $\text{Var} = 1/l^2$
 memoryless: $P(T > s + t | T > s) = P(T > t)$
 if the question gives the MEAN μ instead, $l = 1/\mu$ and $f = (1/\mu) e^{-t/\mu}$

G. Sampling, SE, CLT

SE of the mean $\sigma_{\bar{X}} = \sigma / \sqrt{n}$
 SE estimate $S / \sqrt{n}$
 SE of a proportion $\sqrt{p(1-p)/n}$
 CLT as n grows: $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, under the theorem's conditions
 $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$
 exact, not approx, when the population itself is normal (any n)
 $n \geq 30$ is the course rule of thumb, not a universal theorem cutoff
 for small n and a non-normal population, normal approximation needs justification
 sum of n draws $\sim N(n\mu, n\sigma^2)$
 quadruple $n \rightarrow$ halve the SE (sqrt rule)

H. Estimation

estimator = a rule (a random variable), estimate = the number	
unbiased	$E(\hat{\theta}) = \theta$
consistent	converges in probability; sufficient pair $E \rightarrow \theta$, $\text{Var} \rightarrow 0$
efficient	smallest variance AMONG UNBIASED
relative efficiency of T_1 vs $T_2 = \text{Var}(T_2)/\text{Var}(T_1)$	
sufficient	conditional sample distribution given the statistic is parameter-free;
Neyman-Fisher factorisation is the usual test in this course	
$\text{MSE} = \text{Var} + \text{Bias}^2$	
$\bar{X}$: unbiased, $\text{Var} = \sigma^2/n$, consistent	
$S^2 = \sum (X_i - \bar{X})^2 / (n-1)$: unbiased for σ^2	(the $n-1$ matters)
$\hat{p} = X/n$: unbiased, $\text{Var} = p(1-p)/n$	

I. The four sentence patterns that carry most marks

1	"state which estimator is unbiased"	check E of each, sum the coefficients
2	"find the constant that makes it unbiased" solve sum of coefficients = divisor	
3	"which is most efficient"	variances, smallest among the unbiased
4	"show $\bar{X}$ is consistent"	$E = \mu$ for all n , $\text{Var} = \sigma^2/n \rightarrow 0$

J. Boundaries

in scope	lectures 1 to 21, the four MTE papers plus their schemes, the in-scope parts of
the 2024-25 and 2025-26 assignments, all four S&P decks, the Chebyshev deck	
boundary	the confidence interval shape (named in lecture 19, mechanics in lecture 25)
out	MLE, method of moments, hypothesis testing, t/F/chi-square, ANOVA,
confidence interval mechanics, lms-theory p031-p040	